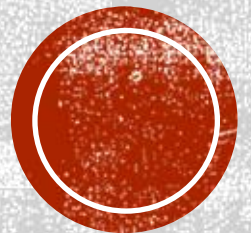




UNDERSTANDING USED CAR PRICES

Prepared by : Zainab Moosa





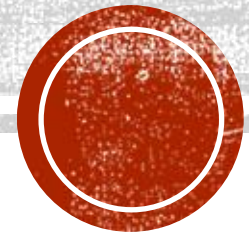
Introduction :

Buying a used car has become a popular choice because it offers more affordable options than buying a new car. However, used cars differ in many aspects, such as manufacturing year, mileage, condition, manufacturer, and vehicle type, which can affect their prices. In this project, a dataset of 426,880 used car listings from the United States will be analyzed to better understand the factors that influence used car prices.



Problem Statement :

With 426,880 used car listings across the United States, buyers have many different options with different prices and specifications. This makes it difficult to understand what affects used car prices and to compare similar cars to determine a fair market price. As a result, buyers may make poor purchasing decisions.



	manufacturer	model	year	price	odometer	condition	transmission	type	size
331947	jeep	patriot	2016	500	154119.0	excellent	automatic	sedan	mid-size
30486	jeep	wrangler unlimited	2016	65000	42600.0	like new	automatic	SUV	full-size

	manufacturer	model	year	price	odometer	condition	transmission	type	size
89975	honda	accord	2016	1000	76000.0	excellent	automatic	sedan	mid-size
320064	honda	odyssey touring elite	2016	49995	21795.0	excellent	automatic	mini-van	full-size

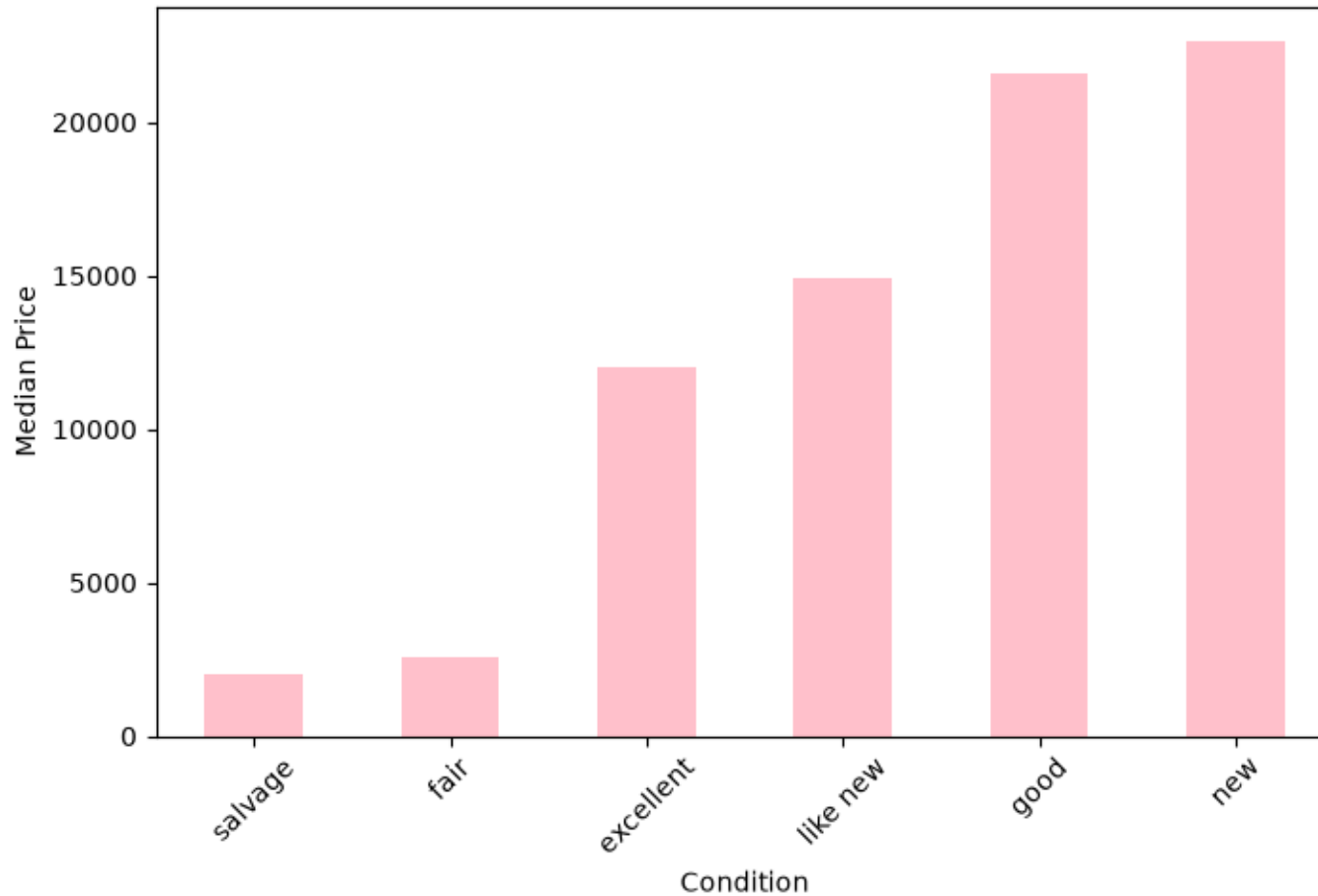
	manufacturer	model	year	price	odometer	condition	transmission	type	size
290389	bmw	530i m5	2001	550	190000.0	fair	automatic	sedan	mid-size
364025	bmw	3 series	2001	15500	95770.0	excellent	manual	convertible	mid-size

Key Finding :

Even cars from the same manufacturer and year can have different prices because of factors such as mileage, condition, transmission, and vehicle type.

OBJECTIVE 1
HOW CAN TWO CARS FROM THE SAME MANUFACTURER AND YEAR HAVE DIFFERENT PRICES?



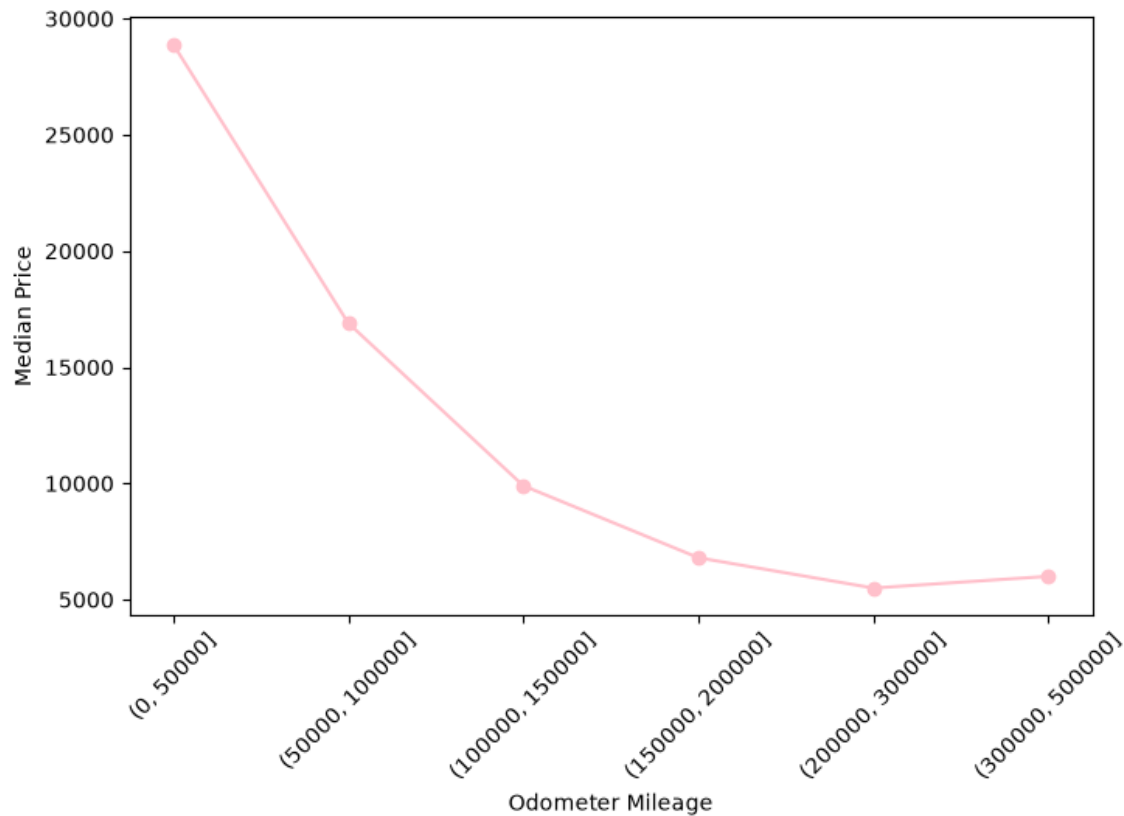


Key Finding :

Used car prices vary depending on the vehicle condition, and cars in better condition generally have higher prices.

OBJECTIVE 2
HOW DO USED
CAR PRICES
VARY ACROSS
DIFFERENT
VEHICLE
CONDITIONS?



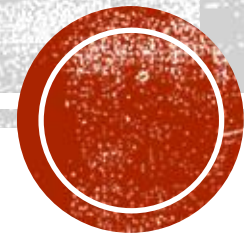


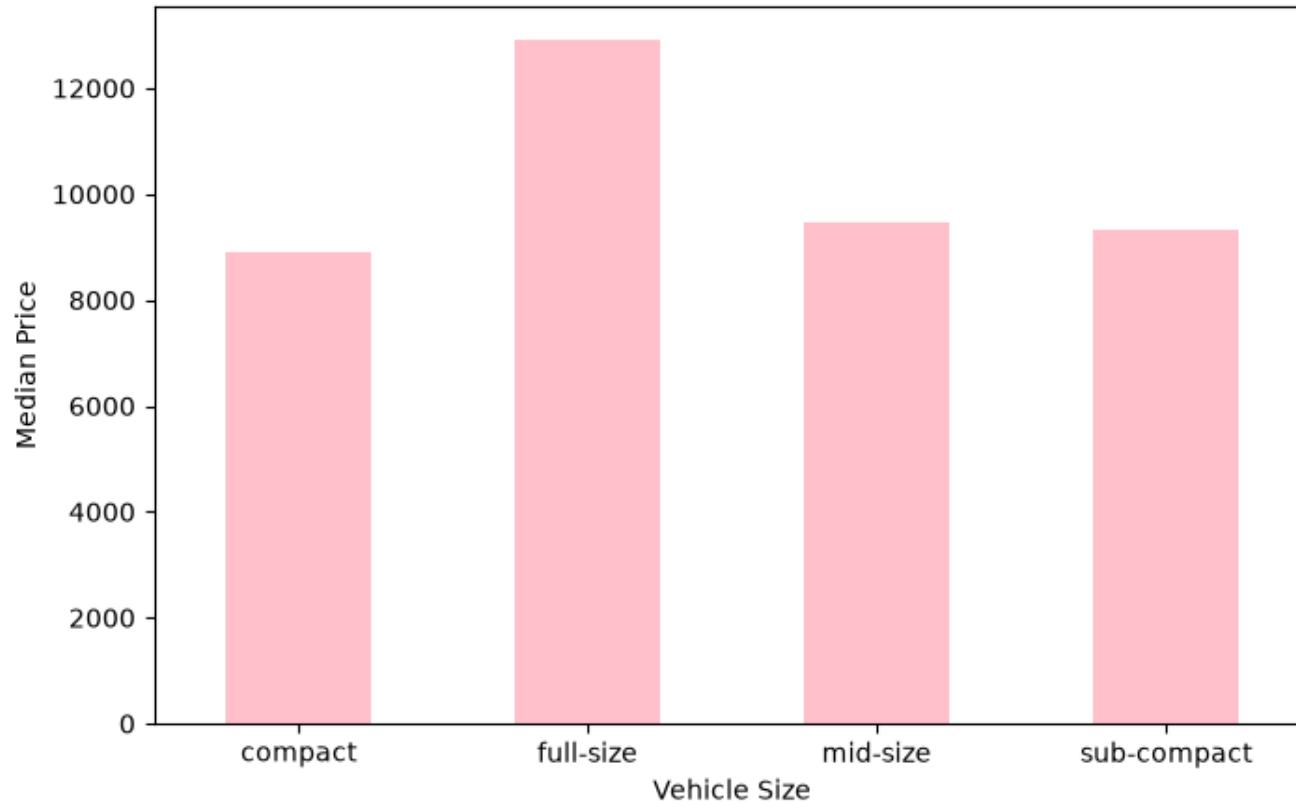
OBJECTIVE 3

HOW DOES ODOMETER MILEAGE AFFECT VEHICLE PRICES?

Key Finding :

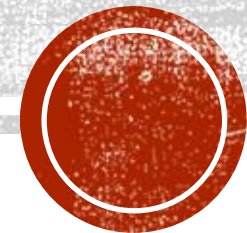
Odometer mileage has a strong effect on price. Cars with lower mileage generally have higher prices.





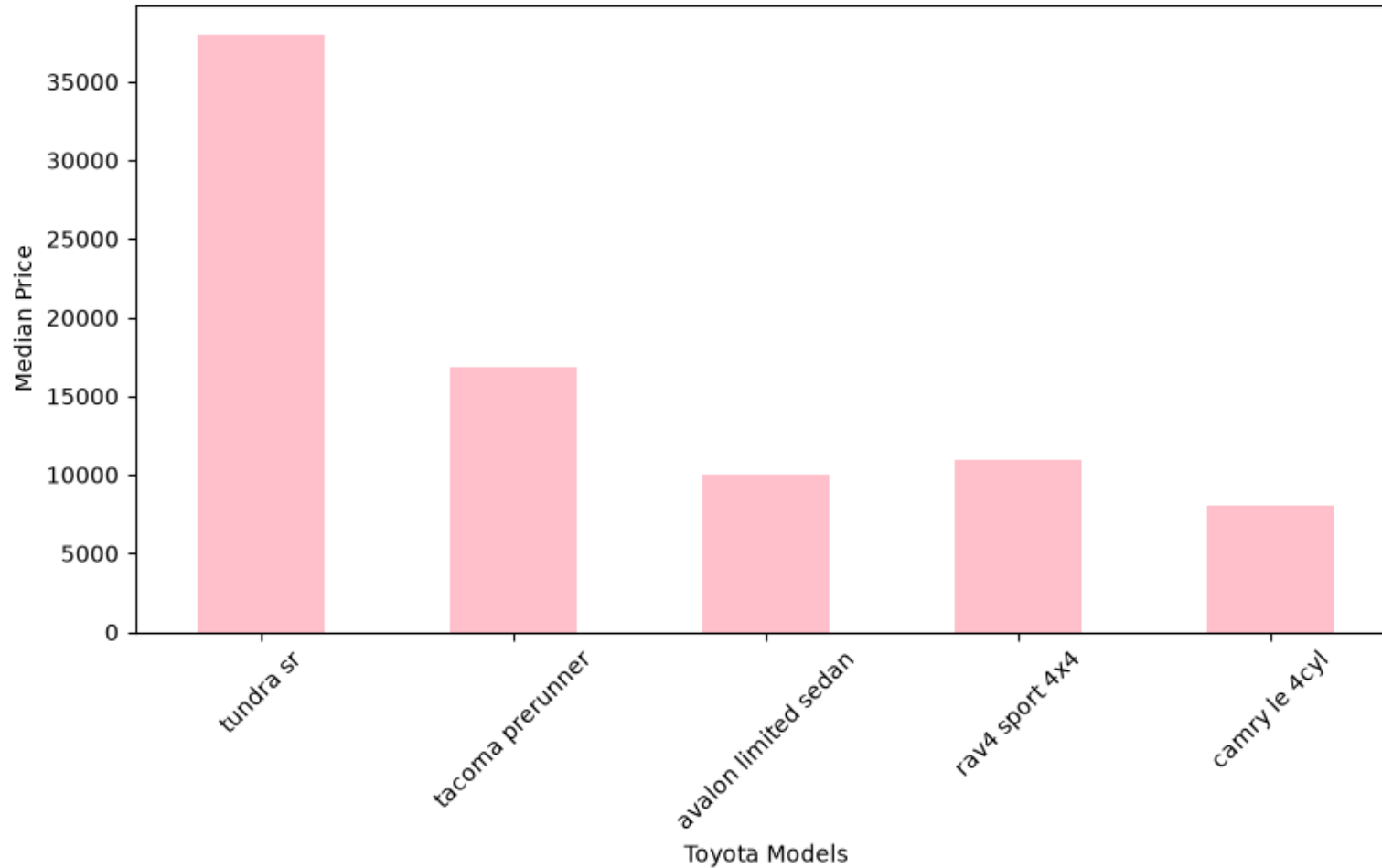
OBJECTIVE 4

**HOW DO USED
CAR PRICES VARY
ACROSS
DIFFERENT
VEHICLE SIZES
AND MODELS?**



Key Finding :

Vehicle size and model affect used car prices. Full-size vehicles generally have higher prices, and some vehicle models are more expensive than others.

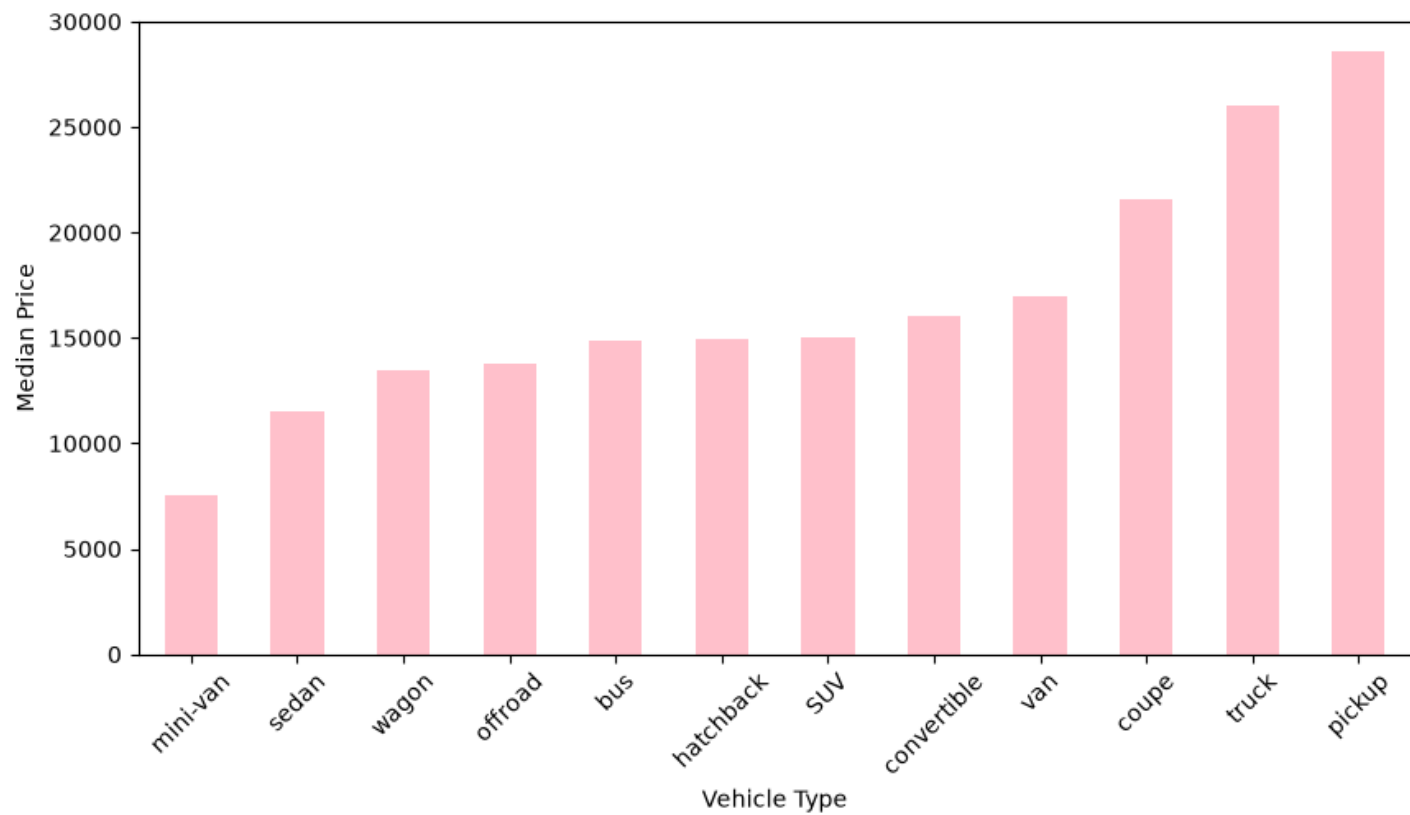


Key Finding :

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OBJECTIVE 4
**HOW DO USED
CAR PRICES VARY
ACROSS
DIFFERENT
VEHICLE SIZES
AND MODELS?**

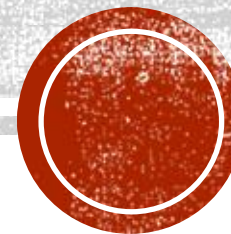


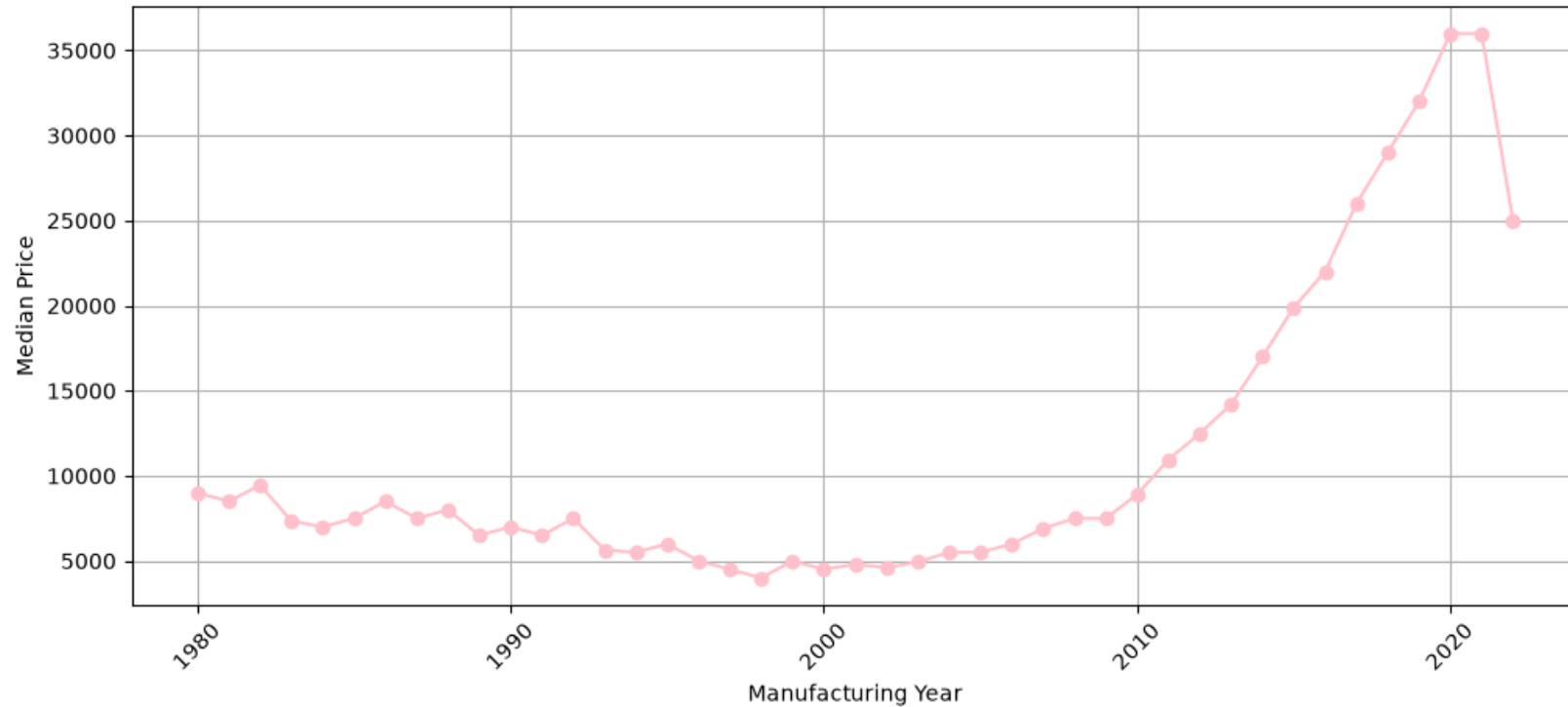


OBJECTIVE 5
HOW DO USED
CAR PRICES
VARY ACROSS
DIFFERENT
VEHICLE
TYPES?

Key Finding :

Vehicle type also has a noticeable effect on used car prices.





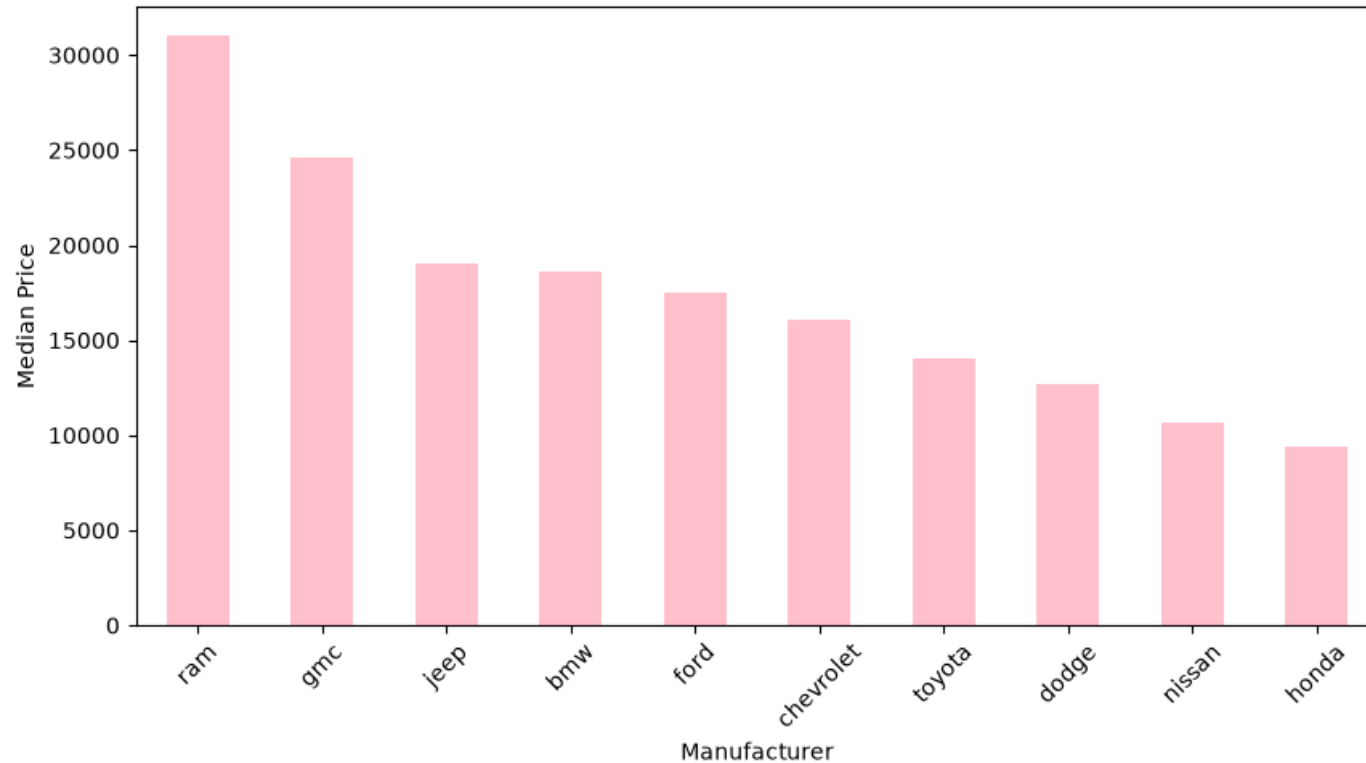
OBJECTIVE 6

HOW DOES THE MANUFACTURING YEAR AFFECT VEHICLE PRICES?

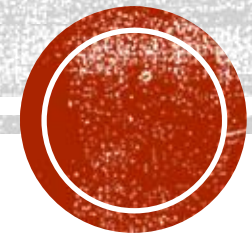
Key Finding :

Prices show a slight increase for very old vehicles, remain lower for some middle-year models, then increase significantly for newer vehicles. However, prices slightly decrease for the 2022 model year .



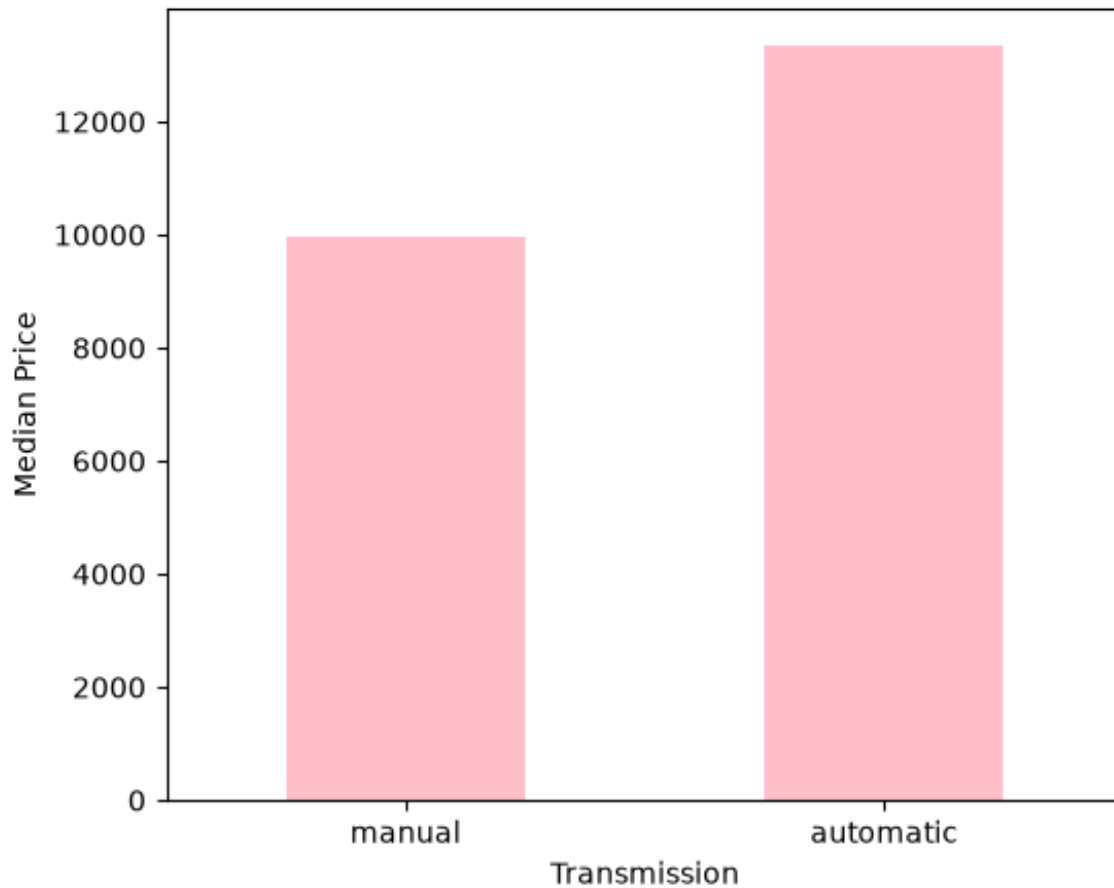


OBJECTIVE 7
HOW DO USED CAR
PRICES VARY
ACROSS
DIFFERENT
MANUFACTURERS?



Key Finding :

The manufacturer has a strong effect on used car prices.

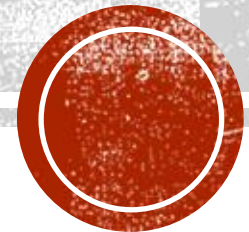


OBJECTIVE 8

**HOW DO PRICES VARY
BETWEEN AUTOMATIC
AND MANUAL
VEHICLES?**

Key Finding :

Transmission type has some effect on price, but the difference between automatic and manual vehicles is not very large.





RECOMMENDATIONS :

- Do not rely only on price when choosing a used car.
- Compare similar cars before making a purchase.
- Consider the manufacturing year and mileage when comparing vehicles.
- Check the vehicle condition carefully.
- Compare prices across different manufacturers.

